A set $F[x] = \{ \sum_{i=0}^{n} a_i \cdot x^i \mid n \in \mathbb{N},\ a_i \in F \}$ with addition and scalar-multiplication is called the polynomial space. As we know, it is countably infinite dimensional,

$$F[x] = \mathrm{Span}_F \{ x^k \mid k \in \mathbb{N} \cup \{0\} \}.$$

In this paper, we investigate some typical properties of the polynomial space, using our background knowledge about dual, normal form, and inner product.

In $\mathbb{C}[x]$, the map

$$\langle \cdot, \cdot \rangle : \mathbb{C}[x] \times \mathbb{C}[x] \to \mathbb{C} : \int_I f(x) \cdot \overline{g(x)}\, dx \quad \text{where } I \text{ is closed interval}$$

is an inner product on $\mathbb{C}[x]$, because: given $f, h, g \in \mathbb{C}[x]$, and $c \in \mathbb{C}$,

$\int_I [cf+h] \cdot \overline{g} = c\int_I f \cdot \overline{g} + \int_I h \cdot \overline{g}$ by the linearity of integral

And

$\overline{\int_I f \cdot \overline{g}} = \int_I \overline{f \cdot \overline{g}} = \int_I \overline{f} \cdot g = \int_I g \cdot \overline{f}$, because

$\overline{\int_I f} = \overline{\int_I \mathrm{Re}f + i \cdot \int_I \mathrm{Im}f} = \int_I \mathrm{Re}f - i \cdot \int_I \mathrm{Im}f = \int_I \mathrm{Re}f - i \cdot \mathrm{Im}f = \int_I \overline{f}$.

Finally

$\int_I f \cdot \overline{f} = \int_I \|f\|^2 \geq 0$. So, shown. □

Def. Let $V = \mathbb{C}[x]$, the space of complex coefficient polynomials.
Define an inner product on V as:

$$\langle f, g \rangle = \int_0^1 f(x)\overline{g(x)}\,dx, \text{ where } \overline{g} \text{ is obtained by take conjugate to coeff. of } g.$$

1. V does not satisfy the Riesz-representation Theorem.

Proof. Let $z \in \mathbb{C}$ be fixed. Let $L: V \to \mathbb{C}: f \mapsto f(z)$ be the evaluation at z.
Clearly, L is a linear functional. Suppose that for some $g \in V$ s.t

$$\langle f, g \rangle = L(f) = f(z) \text{ for all } f \in V.$$

Take $h(x) = x - z$. Then, for any $f \in V$,

$$\langle h \cdot f, g \rangle = \int_0^1 h(t) \cdot f(t) \cdot \overline{g(t)}\,dt = h(z) \cdot g(z) = 0,$$

But $f = \overline{h}\,g$, then

$$\int_0^1 |h(x)|^2 |g(x)|^2 dt = 0, \text{ so } h(x)g(x) = 0, \text{ but } h(x) \neq 0, \text{ s. } g \equiv 0.$$

This implies $L = 0$, but it is contradiction.

2. Adjoint of a linear operator on V may not exist.

Finite subspace of $F[x]$.

Let $P_2(\mathbb{R}) = \text{span}_{\mathbb{R}}\{1, x, x^2\}$, equipped inner product

$$\langle f, g\rangle = \int_0^1 f(t)\cdot g(t)\,dt.$$

1). For fixed $a\in\mathbb{R}$, let $V_a : P_2(\mathbb{R})\to\mathbb{R} : f\mapsto f(a)$.
Find a polynomial $h\in P_2(\mathbb{R})$ s.t

$$\langle f, h\rangle = V_a(f) = f(a) \quad \text{for all } f\in P_2(\mathbb{R}).$$

Proof. Construct an orthonormal basis for $P_2(\mathbb{R})$ as:

$\alpha_1 = 1$.

$\alpha_2 = x - \frac{\langle x, 1\rangle}{\|1\|^2}\cdot 1 = x - \frac{1}{2}$.

$\alpha_3 = x^2 - \frac{\langle x^2, 1\rangle}{\|1\|^2}\cdot 1 - \frac{\langle x^2, x-\frac{1}{2}\rangle}{\|x-\frac{1}{2}\|^2}\cdot(x-\frac{1}{2}) = x^2 - \frac{1}{3} - \cdot(x-\frac{1}{2})$

$= x^2 - x + \frac{1}{6}$.

$\langle x^2, x-\frac{1}{2}\rangle = \int_0^1 x^3 - \frac{1}{2}x^2\,dx = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$.

$\langle x-\frac{1}{2}, x-\frac{1}{2}\rangle = \int_0^1 (x-\frac{1}{2})^2\,dx = 2\cdot\int_0^{\frac{1}{2}} x^2\,dx = 2\cdot\frac{1}{3}\cdot\frac{1}{8} = \frac{1}{12}$

$\langle x^2-x+\frac{1}{6}, x^2-x+\frac{1}{6}\rangle = \int_0^1 x^4 - x^3 + \frac{1}{6}x^2 - x^3 + x^2 - \frac{1}{6}x + \frac{1}{6}x^2 - \frac{1}{6}x + \frac{1}{36}\,dx$

$= \int_0^1 x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36}\,dx$

$= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{8} + \frac{1}{36} = \frac{1}{180}$.

Therefore, $\{1, \sqrt{12}(x-\frac{1}{2}), \sqrt{180}\cdot(x^2-x+\frac{1}{6})\}$ is an orthonormal basis.

Now, $h(x) = \overline{1}\cdot 1 + \overline{\sqrt{12}\cdot(a-\frac{1}{2})}\cdot\sqrt{12}(x-\frac{1}{12}) + \overline{\sqrt{180}\cdot(a^2-a+\frac{1}{6})}\cdot\sqrt{180}(x^2-x+\frac{1}{6})$.

$= 1 + (12a-6)(x-\frac{1}{12}) + 180(a^2-a+\frac{1}{6})(x^2-x+\frac{1}{6})$

is that desired.